Permanent Magnet Synchronous and Brushless DC Motor Drives

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To
My brothers: Govindan and Elango
My sisters: Jothi, Parvathi, and Saya

Contents

PART II Permanent Magnet Synchronous Machines and Their Control

Preface

Books on permanent magnet (PM) ac machine drives have focused primarily on the design of the machines and have covered the control and converters for these drives only in an elementary manner. In the last two decades, research and development on control strategies and their subsequent implementation have been reported in journal publications and presented at conferences. The knowledge base of these publications and conference proceedings is yet to be cohesively made available in a book through a systematic dissemination to both industrial and academic audiences. With regard to power electronics associated with these drive systems, the three-phase bridge inverter has been used as the standard for a very long time. Its understanding and control have changed significantly over time. Cost minimization has been the main focus of emerging high-volume applications and has therefore necessitated a review of the subsystem costs. While controller costs have been standardized according to their applications, the obvious targets for cost minimization are the converters and the machines. Of late, new power converter topologies are also being considered for low-cost drive systems. A book dedicated exclusively to PM ac machines with a major emphasis on the control and low-cost converter topologies would be timely. Keeping this in mind, this book has been under development for several years. The contents of this book were widely used for teaching at the doctoral level at Virginia Tech and tested at the University of Aalborg, Denmark, and industrial audiences in the United States and other countries.

This book is divided into three parts. Part I covers the fundamentals of the machines, power devices, inverters, and their control (Chapters 1 and 2). Parts II and III are devoted to the PM synchronous (Chapters 3 through 8) and brushless dc motor drives (Chapters 9 through 14), respectively.

To understand PM ac drives one must start with the fundamentals of the machine. Chapter 1 provides an elementary introduction to the characteristics of PMs and their operating points, machine rotor configurations, differences between synchronous and brushless dc machines, windings, flux density distribution in the tooth and yoke, expressions relating machine dimensions, magnet and stator excitation with torque and power output, and inductance derivations. Further, the core losses, their computation and measurement are included to cover the characterization of the machine for modeling and control strategy formulations. Greater emphasis has been placed on sinusoidal machines than on trapezoidal machines as the former are closely related to other ac machines both in functioning and in the principles of operation and control. Considering only the fundamental components of the variables, both the sinusoidal and trapezoidal PM machines are identical in their behavior. The applicability of the principles is obvious even though there are some significant differences among them. Various aspects of PM placement on the rotor, special machines with PM on the stator, and the Halbach arrangement are presented to enable readers to familiarize themselves with both the old and the new machines.

Chapter 2 contains a brief description of power devices, their switching characteristics and losses, rectifiers, and inverters. The section on inverters introduces various switching schemes, their merits and demerits, and their modeling. A description of the four-quadrant operation and its applicability to PM ac machines is included to introduce the terminology that is used frequently in the description and operation of PM ac drive systems.

The dynamic modeling of PM synchronous machines using dqo-axes in rotor reference frames and also in space phasor variables is systematically derived in Chapter 3. The approach in this chapter is intuitive and assumes that the reader is familiar with elementary circuit theory and simple axes projections using trigonometry. Various aspects of the modeling are illustrated with MATLAB®-based simulation results. The dynamic model is required in subsequent developments.

The starting basis for the control of PM synchronous machines is vector control, leading to the decoupling of torque and flux channels in the machine. Additional performance requirements such as constant torque angle, unity power factor, constant mutual flux linkages, maximum torque per unit current, fixed angle between current and flux phasors, constant power loss, and maximum efficiency are superimposed on the vector controller and thus various control strategies emerge for the synchronous machine. All these control strategies and some aspects of their implementation for the PM synchronous machines are elaborated in Chapter 4.

Flux weakening is crucial to extend the speed range of PM synchronous motor drives. Various control strategies to accomplish flux weakening are provided in Chapter 5 along with implementations. Importantly, the six-step inverter operation in the flux-weakening mode is studied and compared to other methods of flux weakening based on model and model-free controllers.

The differences between interior and surface PM machines related to their performance in flux weakening as well as in the constant torque regions are derived with normalized machine parameters and are discussed.

The drives have invariably inner current control loops for torque control and hence the necessity for current controllers and their design procedure in practice. Such a procedure is derived from the transfer function obtained from block diagram reduction methods. The method establishes the similarity between the separately excited dc machine and PM synchronous machine drives, thus broadening the reader's understanding. Speed control is enforced with an outer speed feedback control loop. The design of the speed controller is analytically derived using the symmetric optimum method. The design procedure is illustrated with an example in Chapter 6.

Almost all of the control strategies are dependent on machine parameters for their implementation. Because of the sensitivity of the stator resistance, and rotor flux linkages to temperature and q-axis inductance to saturation, the consequences on drive performance and methods to overcome their effects in drive systems are studied in Chapter 7 for PM synchronous machine drives.

Position information is vital to the success of the control of PM synchronous motor drives. The position sensorless approach is very attractive from the point of view of cost and reliability. A few methods of position sensorless control are presented in Chapter 8.

Chapters 9 through 14 contain a study of PM brushless dc motor drives. Chapter 9 elaborates on the modeling and simulation of these drives. A control scheme in prevalent use is also considered in this chapter.

Chapter 10 addresses the important issues of commutation torque ripples and their computation. Flux weakening is studied as an adjunct of the method introduced to study torque generation using Fourier series.

Half-wave inverter topologies emerge to primarily cater to the low-cost and high-performance requirements of high-volume applications. Four such inverter topologies for PM brushless dc motor drives are studied in Chapter 11.

The design of current and speed controllers using a very similar approach taken for PM synchronous motor drives is considered in detail in Chapter 12. A discussion of various methods of current and position sensorless operation of PM brushless dc motor drives is provided in Chapter 13. Various secondary topics such as torque smoothing and parameter sensitivity of these drives are outlined in Chapter 14.

This book is written for engineers with bachelors' degrees, preferably in electrical engineering. The contents of this book can also be taught as a one-semester advanced course in the motor drives area, as practiced in the Electronics and Communication Engineering department, Virginia Tech.

An attempt has been made to develop a consistent method for treating various issues of analysis and design from first principles. While realizing that modeling and analysis get in the way of the flow of the subject matter, they are unavoidable in a book such as this, which is intended for engineers who face the challenges of design and development every day in their work. Wherever appropriate, MATLAB codes are developed and given to illustrate the simulation of the drive systems.

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I have generously drawn on materials from published papers but with my own interpretation, particularly from Dr. G. R. Slemon and his former students, most notably from Dr. T. Sebastian and Dr. X. Liu.

Prof. J. Holtz and his associates' work on inverters, current control, and rotor position sensorless control are gratefully acknowledged.

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The errors in this book are mine while the results of research, on which this text is built, belong to the general community of hundreds of researchers and practitioners in this field.

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Blacksburg, Virginia

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Author

R. Krishnan is the author of *Electric Motor Drives*, published by Prentice Hall in February 2001; its Chinese translation and an Indian edition were released in 2002 by Pearson Education, Taiwan, and Prentice Hall, India, respectively, and its international edition was released in 2001 by Prentice Hall (International Edition). He is also the author of *Switched Reluctance Motor Drives* (CRC Press, first edition, June, 2001; second edition, 2003) and the coeditor and coauthor of *Control in Power Electronics* (Academic Press, August, 2002).

Krishnan has been granted seven U.S. patents and many are still pending in the United States, Europe, and in other countries. His inventions have been prominently featured in the media including radio, TV, and newspapers such as *The Wall Street Journal*. He has served as a consultant for 18 companies in the United States. He has developed and delivered short courses for academia and industry on vector-controlled induction, permanent magnet synchronous and brushless dc, switched reluctance, and linear electric motor drive systems.

Krishnan is a recipient of the best paper prize awards from the IEEE Industry Applications Society's Industrial Drives Committee (five awards) and the Electric Machines Committee (one award). In addition, he received the first prize from the *IEEE Transactions on Industry Applications* for his paper and the 2007 Best Paper Award from the *IEEE Industrial Electronics Magazine*. His coedited book, *Control in Power Electronics*, won the best book award of the Ministry of Education and Sport, Poland, in 2003. He was awarded the IEEE Industrial Electronics Society's Dr. Eugene Mittelmann Achievement Award for Outstanding Technical Contributions to the field of industrial electronics in 2003.

Krishnan is a fellow of the IEEE and a distinguished lecturer of the IEEE Industrial Electronics Society. He is an elected senior AdCom member of the IEEE Industrial Electronics Society and served as a vice president (publications) from 2002 to 2005. He served as the general chair of the 2003 IEEE Industrial Electronics Conference, Roanoke, Virginia, and as one of the three general cochairs of the IEEE IES' ICIT 2006 in Mumbai, India. He has also delivered many keynote speeches in IEEE conferences.

Krishnan and some of his graduate scholars' inventions constituted founding technologies for three motor drives companies. He founded Panaphase Technologies, LLC, in 2002, which went through a successful merger and acquisition in 2007, and Ramu Inc., in 2008, which specializes in high-volume variable speed motor drives for home appliances, air conditioners, hand tools, wind power and automotive applications. He is also a cofounder of TransNetics, LLC, which specializes in intellectual property for linear motor drives.

Krishnan is a professor of electrical and computer engineering at Virginia Tech, Blacksburg, Virginia, and the director of the Center for Rapid Transit Systems, specializing in linear and rotating motor drives.

List of Symbols

A	State transition matrix
a_c	Area of cross section of armature conductor, m^2
A_c	Area of cross section as seen by the flux per pole, m^2
A_g	Area of cross section of air gap, m^2
A_m	Area of cross section of the magnet, m^2
A_{s1}	Linear stator current density, Amp-conductor/m
B_1	Friction coefficient, $N \cdot m/(rad/s)$
B_d	Flux density at maximum energy density operating point, T, and also direct axis flux density, T
B_g	Air gap flux density, T
B_i	Intrinsic magnet flux density, T
B_ℓ	Load constant, $N \cdot m/(rad/s)$
B_m	Operating magnet flux density, T
B_{m1}	Peak of the fundamental magnet flux density, T
B_{mr}	Rms value of B_{m1}, T
B_n	Normalized friction coefficient, p.u.
B_o	Flux density with magnet excitation but with no external excitation, T
B_p	Maximum magnet flux density very near to demagnetization of the magnets, T
	And also the peak flux density in the stator lamination in core loss section, T
B_{pk}	Peak flux density in the air gap, T
B_r	Remanent flux density, T
B_{rr}	Recoil flux density, T
B_t	Total friction coefficient, N m/(rad/s)
B_{tm}	Maximum tooth flux density, T
B_y	Yoke flux density, T
B_{ym}	Maximum yoke (back iron) flux density, T
b_y	Stator back iron thickness, m
C_f	Filter capacitance, F
C_o	Value of the C-dump capacitor, F
D	Bore diameter or inner stator diameter, m
d	Duty cycle
d^r, q^r	d- and q-axes in rotor reference frames, respectively.
e	Instantaneous induced emf, V
	Also denotes average voltage phasor in Chapter 8, V
E	Steady-state (rms) induced emf and also used as C-dump capacitor voltage, V
e_{as}	Induced emf in phase a (instantaneous), V
E_{as}	Steady-state rms phase a induced emf, V

e_{bs}	Induced emf in b phase (instantaneous), V
e_{cs}	Induced emf in c phase (instantaneous), V
e_{hd}, e_{hq}	Nonfundamental harmonic components of induced emf in d- and q-axes, respectively, V
E_m	Maximum energy density of the magnet, J/m^3
E_m	Maximum induced emf in the flux weakening region, V
e_n	Instantaneous normalized induced emf, p.u.
E_n	Normalized steady state induced emf, p.u.
E_p	Peak induced emf, V
E_p	Peak stator phase voltage in PM brushless dc machine, V
F, F^*	Modified reactive power and its reference, VAR
F	Force generated in the machine, N
F_{as}, F_{bs}, and F_{cs}	Instantaneous a, b, and c phase mmfs, AT
$f_{as}(\theta_r)$	Position-dependent function in the induced emf in a phase
$f_{bs}(\theta_r)$	Position-dependent function in the induced emf in b phase
f_c	Control frequency, and PWM carrier frequency, Hz
$f_{cs}(\theta_r)$	Position-dependent function in the induced emf in c phase
F_d	Force density in the machine, N/m^2 and also the mmf per pole in d-Axis winding, AT
F_e	External source mmf, AT
F_m	Magnet mmf, AT
f_n	Natural frequency, Hz
f_s	Utility supply frequency or stator frequency supply, Hz
F_s	Peak value of the resultant stator mmf per pole, AT
F_{sp}	Safe stator peak mmf per pole (that does not demagnetize the magnets), AT
f_{sn}	Normalized stator frequency, p.u.
f_r	Rotational frequency, Hz
g_d	Effective air gap length, m
g_d, g_q	Effective air gap length in direct (magnet) and quadrature axis, m
h	Average duty cycle of phase switches in half-wave converter topologies
H	Inertial constant in normalized unit, p.u.
H	Magnet field intensity, A/m
H_c	Gain of the current transducer, V/A
H_c	Magnet coercive force or coercivity at normal flux density, A/m
H_{ch}	Maximum safe field intensity without demagnetizing the magnets, A/m
H_{ci}	Intrinsic coercive force (where intrinsic flux density is zero) for magnets, A/m
H_{cl}	Maximum field intensity that will not irreversibly demagnetize the low-grade magnets, A/m
H_{cr}	Fictitious coercivity corresponding to remanent flux density of the low-grade magnets, A/m
H_{cs}	Fictitious coercivity corresponding to recoil remanent flux density of the low-grade magnets, A/m

H_e	External field intensity, A/m
H_g	Air gap field intensity, A/m
H_m	Magnet field intensity, A/m
H_o	Magnet field intensity with zero external excitation and on no load, A/m
H_p	Peak field intensity without demagnetizing the magnets, A/m
H_ω	Gain of the speed filter, V/(rad/s)
i_α, i_β	Currents in α and β frames that are rotating at assumed speed, A
i_{abc}	abc phase current vector
i_{as}	Instantaneous stator phase a current, A
i_{asn}	Normalized phase a stator current (instantaneous), p.u.
I_{avn}	Normalized average armature current, A
I_b	Base current in two-phase system, A or base current, A
I_{b3}	Base current in three-phase system, A
i_{bs}, i_{cs}	Instantaneous b and c stator phase currents, A
I_{cn}	*n*th harmonic capacitor current, A
I_d	Steady-state d-axis current flowing into the motional emf in rotor reference frames, A
I_{dc}	Steady-state dc link current or the inverter input current, A
i_{ds}, i_{qs}	d- and q-axes stator currents in stator reference frames, A
i_{dsi}, i_{qsi}	d- and q-axes stator currents in stator reference frames due to injected high-frequency voltage signals, A
i^s_{dsi}, i^s_{qsi}	d- and q-axes stator currents in stator reference frames due to injected high-frequency voltage signals, A
i^r_{dsi}, i^r_{qsi}	d- and q-axes stator currents in rotor reference frames due to injected high-frequency voltage signals, A
I_f	Steady-state field current or flux producing component of the stator current phasor, A
i_f	Instantaneous field current or flux producing component of the stator current phasor, A
i_{hk}	Harmonic current vector, A
i_{ip}, i_{in}	Positive and negative sequence current components resulting from injected signals, A
I_{me}	Rms per phase equivalent magnet current, A
I_{mep}	Peak per phase equivalent magnet current, A
I_n	*n*th harmonic current, A
i_o	Stator zero sequence current, A
I_p	Peak stator phase current in PM brushless dc machine, A
I_{ph}	Fundamental phase current, A
I_{ps}	Peak stator phase current in PM synchronous machine, A
I_q	Steady-state q-axis current flowing into the motional emf in rotor reference frames, A
I_{qc}, I_{dc}	Core loss current in the stator d- and q-axes, A
i_{qdo}	qdo current vector (q, d, and o) in stator reference frames, A
i^r_{qdo}	qdo current vector (q, d, and o) in rotor reference frames, A
i_{qds}	Stator q- and d-axes current vector in stator frames, A

i^{r}_{qds}	Stator q- and d-axes current vector in rotor reference frames, A
I_{qs}, I_{ds}	Steady-state stator q- and d-axes currents in stator reference frames, A
I^{r}_{qs}, I^{r}_{ds}	Steady-state stator q- and d-axes currents in rotor reference frames, A
I_s	Source current, A
I_{sm}	Maximum safe stator current that does not demagnetize the magnets, A Also denotes fundamental value of the stator current phasor under SSV strategy, A
I_{sp}	Sinusoidal peak stator phase current, A
I_{sy}	Rms current of the synchronous machine phase, A
i_T	Torque producing component of the stator current phasor, A
i^*_f	Field current reference, A
i_T	Torque producing component of stator current phasor, A
i^*_T	Reference torque producing component of stator current phasor, A
i^r_s	Stator current phasor in rotor reference frames
i^s_{sip}, i^s_{sin}	Positive and negative sequence current phasor components resulting from injected signals, A
I_T, I^*_T	Steady-state values of i_T, and i^*_T
$i^*_{as}, i^*_{bs}, i^*_{cs}$	a, b, c phase current commands, A
i^r_{xsn}	Current in rotor reference frames, and subscript x for q- or d-axis, s for stator, and n for normalized unit. Without subscript n, the variable is in SI units.
J	Total moment of inertia, kg-m^2
J_c	Current density of the armature conductor, A/m^2
J_l	Moment of inertia of load, kg-m^2
J_m	Moment of inertia of motor, kg-m^2
k	Inverse of leakage factor
K_b	Induced emf constant, V/(rad/s)
K_c	Gain of the current controlle, V/A
k_d	Winding distribution factor for the fundamental
k_{dn}	Winding distribution factor for the *n*th harmonic
k_e	Eddy current loss constant
k_h	Hysteresis loss constant
K_i	Current loop transfer function gain
K_{is}	Integral gain of the speed controller
k_m	Ratio between mutual and self-inductances, per phase, in PMBDC machine
k_p	Winding pitch factor for the fundamental
k_{pn}	Winding pitch factor for the *n*th harmonic
K_i	Integral gain of the PI controller in power feedback control
K_p	Proportional gain of the PI controller in power feedback control
K_{ps}	Proportional gain of the speed controller
K_r	Inverter gain, V/V

K_s	Gain of the speed controller
k_s	Ratio of slot to tooth width
k_{sk}	Skew factor
K_t	Torque constant, N m/A
K_{vf}	Ratio between stator phase voltage and stator frequency, V/Hz
k_ω	Fundamental winding factor
K_ω	Speed feedback filter gain, V/(rad/s)
$k_{\omega n}$	nth harmonic winding factor
L	Stack length, m
L, M	Stator self and mutual inductances in PM brushless machines, H
L_a	Denotes (L-M) in PM brushless dc machines, H
L_{aa}	Self-inductance of a phase winding, H
L_{ab}	Mutual inductance between phases a and b, H
L_{ac}	Mutual inductance between phases a and c, H
L_b	Base inductance, H
L_f	Filter inductance, H
L_{ma}	Magnetizing inductance per phase, H
ℓ_c	Average length of a stator phase winding, m
ℓ_g	Air gap length, m
ℓ_m	Magnet length, m
L_o	Value of inductor in energy recovery chopper in C-dump topology, H Also denotes zero sequence inductance in machines, H
L_q, L_d	Quadrature and direct axis stator self-inductances in rotor reference frames, H
L_{dq}	Mutual inductance between d- and q-axes, H
L_{qd}	Mutual inductance between q- and d-axes, H
L_{ql}, L_{dl}	Stator q- and d-axes leakage inductances in rotor reference frames, H
L_{qn}, L_{dn}	Normalized quadrature and direct axis stator self-inductances in rotor reference frames, p.u.
L_{qq}, L_{dd}	Self-inductance of the stator q- and d-axes stator windings in stator reference frames, H
L_s	Input source inductance, H
L_{xy}	Mutual inductance between windings given by the subscripts x and y, H
m	Modulation ratio and also used to denote the ratio between slot to pole number
N_1	Number of turns/phase with half-wave converter control
N_{co}	Cogging cycles per mechanical revolution
N_{ph}	Effective turns per phase
n_r	Rotor speed, rpm
n_s	Stator field speed or synchronous speed, rpm
N_{sp}	Number of slots per pole (and also denotes speed in rpm in Chapter 2)

o	Subscript ending with o in a variable indicates its steady-state operating point value
p	Differential operator, d/dt
P	Number of poles
P_1	Air gap power with half wave converter, rad/s
P_{1n}	Dominant harmonic resistive losses, W
P_a	Air gap power, W
P_{an}	Normalized air gap power, p.u.
P_{av}	Average input power, W
P_b	Base power, W
P_c	Armature resistive losses, W
P_{c1}	Machine copper loss with half-wave converter operation, W
P_{co}	Core losses, W
P_{cd}	Core loss density (P_{ed} + P_{hd}), W/unit weight
P_{ed}	Eddy current loss per unit weight, W/unit weight
P_{et}	Eddy current loss in tooth for trapezoidal magnet flux density distribution, W
P_{ets}	Eddy current loss in tooth for sinusoidal magnet flux density distribution, W
P_{ey}	Eddy current loss in yoke for trapezoidal magnet flux density distribution, W
P_{eys}	Eddy current loss in yoke for sinusoidal magnet flux density distribution, W
P_{eyn}	Ratio of P_{ey} and P_{eys}
P_{hd}	Hysteresis current loss per unit weight, W/unit weight
P_{hs}	Stator hysteresis loss, W
P_i	Input power, W
P_p	Number of pole pairs
p_i	Instantaneous input power, W
P_l	Total power losses, W
P_{lm}	Maximum power losses, W
P_m	Mechanical power output, W
P_o	Output power, W
P_{on}	Normalized power output, p.u.
P_{sc}	Stator resistive losses of a three phase machine, W and, also the conduction loss in a transistor, W
P_{scn}	Normalized stator resistive losses of a three phase machine, p.u.
P_{sw}	Switching loss in a device, W
P_{VA}	Apparent power, VA
P_a^*	Air gap power reference, W
q	Number of slots per pole per phase
Q and Q_i	Reactive power, VAR
Q_f	Filtered reactive power, VAR
Q_n, Q_{fn}	Normalized reactive and filtered reactive power, respectively, p.u.
r	Radius of the bore, m
R_1	Stator resistance/phase with half-wave converter operation, W

R_a	Twice the per phase resistance in PM brushless dc machines, Ω
R_c	Core loss resistance, Ω
R_d, R_q	Stator d- and q-axes winding resistances, Ω
R_s	Stator resistance per phase, Ω
R_{sn}	Normalized stator resistance per phase, p.u.
s	Laplace operator
S	Number of slots in the stator
S_a, S_b, S_c	a, b, c phase switching states of the inverter
T	Carrier period time, s, as well as effective turns per stator phase winding
t	Time, s
t_d	Inverter dead time, s
t_t	Time taken to vary the tooth flux density from zero to maximum value, s
t_y	Time taken to vary the yoke flux density from zero to maximum value, s
T_1, T_2	Electrical time constants of the motor, s
T_{abc}	Transformation from abc to qdo variables in rotor reference frames
T_{av}	Average torque, N · m
T_b	Base torque, N · m
T_c	Time constant of the current controller, s
T_{co}	Cogging torque for a cycle expressed in terms of a Fourier series, N · m
T_e	Air gap or electromagnetic torque, N · m
T_{ec}	Analytically computed cogging torque, N · m
T_{e1}	First-harmonic air gap torque, N · m
T_{e6}	Sixth-harmonic torque, N · m
T_{e6n}	Normalized sixth-harmonic torque, p.u.
T_{ec}	Torque reference generated by speed error, N · m
T_{ef}	Maximum air gap torque generated with the voltage and current constraints, N · m
T_{em}	mth harmonic torque in PM brushless dc machines, N · m
T_{emn}	Normalized mth harmonic torque in PM brushless dc machines, p.u.
T_{en}	Normalized air gap torque, p.u.
T_{er}	Rated air gap torque, N · m
T_{er}	Reluctance torque, N · m
T_{es}	Synchronous torque due to magnets, N · m
T_i	Time lag of the current control loop, s
T_l	Load torque, N · m
T_{ln}	Normalized load torque, p.u.
T_m	Mechanical time constant, s
T_{max}	Maximum torque limit, N · m
T_n	Peak value of nth order harmonic of cogging torque, N · m
t_{on}	Conduction time or on time of a device in a switching cycle, s

t_{off}	Device off time in a switching cycle, s
T_p	Peak value of cogging torque in a cycle, N·m
T_{ph}	(Effective) Number of turns per phase
T_r	Converter (inverter) time delay, s
$[T^r]$	Transformation from rotor reference frames of the qd variables to stationary reference frames
T_s	Time constant of the speed controller, s
t_s	Total switching time in a device, s
T_ω	Time constant of the speed filter, s
T^s_{abc}	Transformation from abc to qdo variables in the stationary reference frames
T^*_e	Torque reference, N·m
T^*_{en}	Normalized torque reference, p.u.
u	Input vector
V	Rms line-to-line voltage of the supply, V
v_α, v_β	Voltages in α and β frames that are rotating at assumed speed, A
v_{as}, v_{bs}, v_{cs}	Phase a, b, and c input voltages to PM brushless dc machine, V
v_{ab}	Instantaneous line-to-line voltage between phases a and b, V
V_{ab}, V_{bc}, V_{ca}	Rms line-to-line voltages between a and b, b and c, and c and a phases, V
v_{ab}, v_{bc}, v_{ca}	Instantaneous line-to-line voltages between a and b, b and c, and c and a phases, V
v_{abc}	abc voltage vector
v_{abi}	Instantaneous input converter line-to-line voltage between phases a and b, V
v_{am}, v_{bm}, v_{cm}	Instantaneous voltages between dc midpoint voltage and inverter midpole phases a, b, and c, V
v_{an}	Midpoint voltage for ideal inverter phase a, V
v'_{an}	Inverter phase a midpoint voltage with dead time incorporated, V
v_{anc}	Inverter phase a midpoint voltage with dead time compensation, V
v_{ao}, v_{bo}, v_{co}	Inverter midpole voltages, V
V_{as}	Stator rms voltage input per phase and also rms phase a voltage, V
V_{asn}	Normalized phase a stator voltage, p.u.
V_b	Base voltage, V
V_{b3}	Base voltage in three-phase system, V
v_c	Control voltage, V
V_c	Voltage across the capacitor, V
V_{cn}	Normalized conduction voltage drop across a transistor, p.u.
V_{cm}	Maximum control voltage, V
V_d	Diode peak voltage, V
V_{dc}	Steady-state dc link voltage, V
v_{dc}	Instantaneous dc link voltage, V
v_{ds}, v_{qs}	d- and q-axes stator voltages in stator reference frames, V
v^r_{dsi}, v^r_{qsi}	d- and q-axes injected voltages in rotor reference frames, V
V_g	Air gap volume, m^3

v_{g1}, v_{g4}	Gate signals to inverter phase a, V
v'_{g1}, v'_{g4}	Modified gate signals to inverter phase a incorporating dead time, V
v'_{g1c}, v'_{g4c}	Compensated gate signals of inverter phase a to overcome the effects of dead time, V
v_{hk}	Harmonic voltage vector, V
V_i	Inverter input voltage, V
V_m	Peak supply voltage, V and also magnet volume, m^3
v_{mn}	Voltage between midpoint source and neutral of the load, V
v_o	Stator zero sequence voltage, V
V_{on}	Conduction voltage drop across transistor, V
V_{ph}	Maximum rms phase voltage output of the inverter, V
v^r_{qds}	Stator voltage vector in rotor reference frames, V
v_r	Instantaneous phase converter input voltage, V
v_{re}	Instantaneous rectifier output voltage, V
v_s	Instantaneous phase source voltage, V
V^a_{si}	Injected voltage vector in anisotropic or estimated rotor frames, V
V^s_{si}	Injected voltage vector transformed to stator reference frames, V
V^r_{si}	Injected voltage vector transformed to rotor reference frames, V
V_s	Average source voltage, V
V_t	Stator tooth volume, m^3
V_{ts}	Peak power switch voltage, V
V_y	Stator yoke (back iron) volume, m^3
V_{vs}	Sensor output voltage, V
v^*_a	Peak value of the phase command or reference, V
v^r_s	Stator voltage phasor in rotor reference frames
v^r_{sn}	Normalized stator voltage phasor in the rotor reference frame, p.u.
v^r_{xsn}	Voltage in rotor reference frames, and subscript x for q- or d-axis, s for stator and n for normalized unit. Without subscript n, the variable is in SI units.
v_{zs}	Zero sequence voltage, V
W_c	Coenergy, J
W_m	Energy stored in a device, J
W_s	Stator slot width, m
W_t	Stator tooth width, m
X	State variable vector
$X(0)$	Initial steady-state vector
$X(s)$	Laplace transform of input vector
$y(s)$	Laplace transform of output variable
Δi	Hysteresis current window, A
α	Stator voltage phasor angle with reference to rotor d-axis, rad
β	Ratio of actual to controller instrumented mutual inductance and also one-half of the magnet arc, rad
δ	Preceding a variable indicates small-signal variation

δ	Torque angle in synchronous machine, rad
δ^*	Torque angle command in synchronous machine, rad
$\delta\theta$	Error in rotor position, rad
$\delta\omega_m$	Change in rotor speed, rad/s
ΔQ	Change in reactive power, VAR
ϕ	Machine power factor angle, rad
ϕ_i	Input power factor angle, rad
ϕ_g	Air gap flux, Wb
ϕ_m	Peak mutual flux and also magnet flux, Wb
γ	Slot pitch or angle, rad
ξ	Coil pitch, rad
λ	Flux linkage, V-s
λ_{aa}	Flux linkage of a phase winding, V-s
λ_{af}	Armature flux linkages due to rotor magnets, V-s
λ_{af}^*	Armature flux linkages due to rotor magnets at ambient temperature, V-s
λ_{afhd}, λ_{afhq}	Nonfundamental harmonic flux linkages in d- and q-axes, respectively, V-s
λ_{afn}	Normalized mutual flux linkages due to rotor magnets, p.u.
λ_{as}, λ_{bs}, λ_{cs}	Flux linkages of phases a, b and c, respectively, V-s
λ_b	Base flux linkages, V-s
λ_m	Mutual air gap flux linkages, V-s
λ_m^*	Reference or command mutual air gap flux linkages, V-s
λ_{ma}	Flux linkages due to resultant mmf, V-s
λ_{mn}	Normalized mutual flux linkages, p.u.
λ_o	Stator zero sequence flux linkages, V-s
λ_p	Peak mutual flux linkages from rotor magnets (in PM brushless dc machines), V-s
λ_{qs}, λ_{ds}	Stator flux linkages in q- and d-axes, V-s
λ_{qsi}, λ_{dsi}	Stator flux linkages in q- and d-axes for injected high-frequency currents, V-s
λ_s^r	Stator flux linkage phasor in rotor reference frame, V-s
λ_{sn}^r	Normalized stator flux linkage phasor rotor reference frame, p.u.
θ	Spatial position, rad
θ_a	Phase advance angle of current in PM brushless dc machines, rad
θ_{ms}	Angle between the mutual flux and stator current phasors, rad
θ_λ	Angle between stator flux linkages phasor and rotor flux linkages phasor, rad
θ_r	Rotor position, rad
θ_s	Stator voltage phasor angle with respect to the closest switching voltage vector of the inverter, rad Also stator current phasor angle from stationary reference, rad
θ_s^*	Stator current phasor angle command, rad
θ_{sk}	Skew angle, rad
θ_{re}	Estimated rotor position, rad

$\Re$	Reluctance
$\Re_d$, $\Re_q$	Direct and quadrature axis reluctances, respectively
ρ	Saliency ratio, i.e., between q- and d-axes self-inductances and also used to denote specific resistivity of winding material
ρ_i	Mass density of the steel lamination, kg/m^3
τ_{fw}	Stator electrical time constant with full-wave converter operation, s
τ_{hw}	Stator electrical time constant with half-wave converter operation, s
τ_q, τ_d	Stator q- and d-axes stator time constant, s
τ_s	Stator time constant in surface mount PMSM, s
ω_1	Speed with half-wave converters, rad/s
ω_a	Estimated rotor and or anisotropic angular frequency, rad/s
ω_b	Base angular frequency, rad/s
ω_c	Carrier angular frequency, rad/s and also the speed of arbitrary reference frames in induction machines
ω_i	Angular frequency of the injected signal to the stator windings, rad/s
ω_m	Rotor mechanical speed, rad/s
ω_{rn}	Normalized rotor speed, p.u.
ω_{mr}	Speed signal from the output of speed filter, V
ω_r	Electrical rotor speed, rad/s
ω_r^*	Speed reference, V
ω_{rm}	Speed of model rotor frames, rad/s
ω_{rn}	Normalized rotor speed, p.u.
ω_s	Supply angular velocity, rad/s
μ_o	Permeability of air
μ_c	Permeance coefficient of the magnet
μ_{rm}	Relative permeability of the magnet
μ_r, μ_{rec}	Magnet recoil permeability
$\mu_{re}(H_m)$	External permeability of the magnet defined in Chapter 1

Part I

Introduction to Permanent Magnets and Machines and Converters and Control

1 Permanent Magnets and Machines

An introduction to permanent magnets (PMs) and their varieties, their characteristics such as demagnetization, and energy density and realization of machines with PMs is briefly presented in this chapter. Various forms of magnet placement on the rotor lead to certain unique operating characteristics of the machines and a number of such rotor configurations are discussed. PM synchronous machines (PMSMs) with magnets on the stator are of recent origin and they are covered under hybrid machines. The principle of operation of synchronous machines with PM rotors is introduced in the course of which machine fundamental relationships such as the magnetomotive force (mmf), induced emf (also known as back emf), and torque are derived for PMSMs with sinusoidal-induced emf but they can be extended along the same lines to brushless dc machines with trapezoidal-induced emf. To obtain these relationships, it is important to understand the stator windings and how the number of turns and their placement in the stator laminations affect them. Factors that affect the windings are pitch, distribution, and skew, they are evaluated and some commonly used winding types are given. The fundamental machine derivations are made from the first principles and they relate the physical dimensions of the machine and electrical and magnetic loading to output variables such as electromagnetic torque and air gap power [1–8]. In this process, the interplay of the machine variables and control variables is brought to the forefront, demonstrating the sphere of influence for the machine designer and control engineer to influence the outcome of the motor drive performance. In turn, it clearly emphasizes the need for integrated design of the motor drive system to optimize a set of performance indices dictated by a given application. Preliminary sizing of the machine and calculation of motor parameters such as inductances are derived. They are the starting points for fine-tuning the machine design mostly with finite element analysis software, which is outside the scope of this book and hence will not be covered. Core and resistive losses play crucial roles in the power density and operational torque and power of the machine and their computation is explained in detail. A special problem of the PMSMs exists in the form of cogging torque resulting from the interaction of the magnets and the stator teeth. It is described in detail along with the methods for mitigating the cogging torque.

1.1 PERMANENT MAGNETS

Materials to retain magnetism were introduced in electrical machine research in the 1950s [9–14]. There has been a rapid progress in these kinds of materials since then. The magnetic flux density in the magnets can be considered to have two components. One is intrinsic and, therefore, due to the material characteristic depends on the permanent alignment of the crystal domains in an applied field during magnetization. It is referred to as the intrinsic flux density characteristic of the PMs. The flux density component, known as intrinsic flux density, B_i, saturates at some magnetic field intensities and does not increase with the applied magnetic field intensity. The other component of the flux density in the magnet is due to its magnetic field intensity as though the material does not exist in the presence of the applied magnetic field or in other words, is a very small component due to the magnetic field intensity in the coil in vacuum, B_h. Therefore, the flux density in the magnet material is given by

$$B_m = B_h + B_i \tag{1.1}$$

The excitation component B_h is directly proportional to magnetic field intensity, H, and given by

$$B_h = \mu_0 H \tag{1.2}$$

where H is the magnetic field intensity. In all magnetic materials, this component is very small compared to the intrinsic flux density. Combining Equations 1.1 and 1.2, the magnetic flux density can be written as

$$B_m = B_i + \mu_0 H \tag{1.3}$$

where
B_i is the intrinsic flux density
H is the magnetic field intensity
B_m is the flux density in the magnet or known as normal flux density in the magnet

A typical ceramic magnet's intrinsic and magnet flux densities are shown for the second quadrant in Figure 1.1. The magnetic flux density in the second quadrant is a straight line and it can be represented in general as

$$B_m = B_r + \mu_0 \mu_{rm} H \tag{1.4}$$

where μ_{rm} is the relative permeability of the magnet. It is seen that at $H = 0$, the intrinsic and normal induction flux densities pass through the point known as remanent flux density, B_r. The intrinsic flux density in the second quadrant can be derived from the demagnetization characteristic of the magnet as follows:

$$B_i = B_m - \mu_0 H = B_r + \mu_0 H (\mu_{rm} - 1) \tag{1.5}$$

For a hard magnet with a straight line demagnetization characteristic, note that the intrinsic flux density is constant in the second quadrant, i.e., it remains "permanently magnetic" and this is termed a high-grade magnet. If the demagnetization

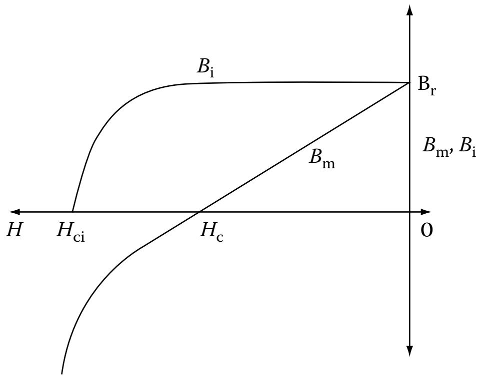


FIGURE 1.1 Typical ceramic magnet flux densities.

characteristic is not a straight line in the second quadrant, then the intrinsic flux density is not a constant, indicating that the "permanency of magnetism" is not as good as that of the high-grade magnets and, therefore, such magnets are known as low-grade PMs. The coercive force required to bring the magnet's intrinsic flux density to zero is the intrinsic coercive force, H_{ci}, and that of the normal flux density is coercive force or coercivity, H_c, shown in Figure 1.1.

The demagnetization characteristic is the one used in analysis and design of the machines, as it includes the effect of the external magnetic field intensity that invariably comes into existence in the machine due to its winding excitation. More on this is described in later sections.

The materials that retained magnetism are known as hard magnet materials. The ability to retain permanent magnetism is found in cobalt, iron, and nickel, and they are referred to as ferromagnetic materials. Various materials such as alnico-5, ferrites (ceramics), samarium-cobalt, and neodymium boron iron are available as PMs for use in machines. The most popular ones in practice are samarium cobalt and neodymium-type magnets.

1.1.1 Demagnetization Characteristics

The generic *B–H* demagnetization characteristics of these materials are shown in Figure 1.2 for second quadrant only [6]. For available PM materials in the market, the reader is referred to manufacturer data sheets that may be obtained from the manufacturer and the data will be available for complete *B–H* characteristics in four quadrants of magnetic flux density versus magnetic field intensity and not limited to second quadrant only. Second quadrant is considered here because the magnets need not have an external excitation once they are magnetized and any magnetic field strength that may be applied through coils in an electromagnetic device is by design usually intended to reduce its flux density. Therefore, the operation of the machine and the magnets lies in the demagnetization domain. That includes the second and third quadrants of the *B–H* characteristics. It is not usual to encounter the third quadrant in operation and hence only the second quadrant is considered here but the analysis and discussion here can be easily extended to cover the third quadrant also, if necessary.

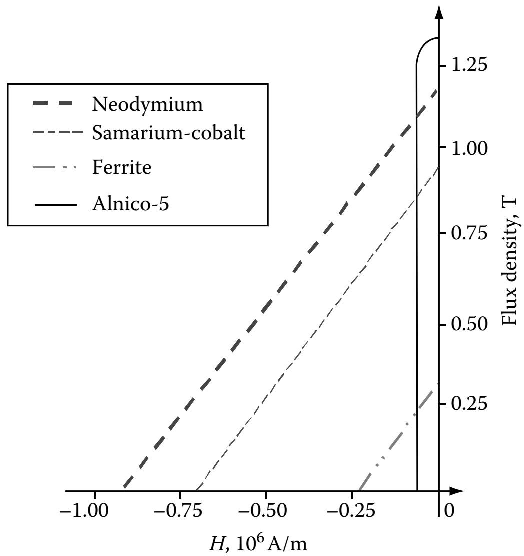


FIGURE 1.2 Second quadrant *B–H* characteristics of PMs. (From Krishnan, R., *Electric Motor Drives*, Figure 9.1, Prentice Hall, Upper Saddle River, NJ, 2001. With permission.)

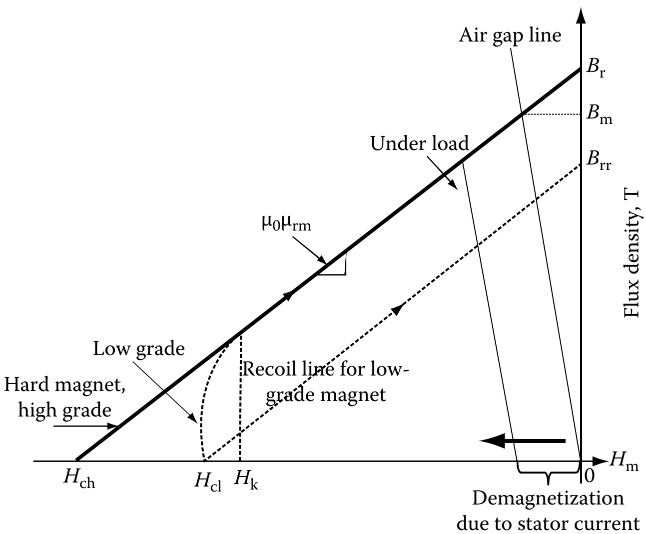


FIGURE 1.3 Operating point of magnets. (From Krishnan, R., *Electric Motor Drives*, Figure 9.2, Prentice Hall, Upper Saddle River, NJ, 2001. With permission.)

The last three listed magnets have straight-line characteristics whereas alnico-5 has the highest remanent flux density but has a nonlinear characteristic. The flux density at zero excitation is known as remanent flux density, B_r (referred to as remanence also in some texts) shown in Figure 1.3 for a generic hard PM. The low-grade magnet has a curve, usually termed as knee at low flux density and around this point, it sharply dives down to zero flux density and reaches the magnetic field strength of H_{cl} known as coercivity. The magnetic field strength at the knee point is H_k. If the external excitation acting against the magnet flux density is removed, then it recovers magnetism along a line parallel to the original *B–H* characteristic. In that process, it reaches a new remanent flux density, B_{rr}, considerably lower than the original remanent flux density. The magnet has irretrievably lost a value equal to $(B_r - B_{rr})$ in its remanent flux density. Even though the recoil line is shown as a straight line, it usually is a loop and the average of the looping flux densities is represented by the straight line. In the case of a high-grade magnet, the *B–H* characteristic is a straight line and its coercivity is indicated by H_{ch}. The line along which the magnet can be demagnetized and restored to magnetism is known as recoil line. The slope of this line is equal to $\mu_0\mu_{rm}$ as derived from the relationship between the flux density and magnetic field intensity and where μ_{rm} is relative recoil permeability for the high-grade magnets. For samarium-cobalt and neodymium boron magnets, the recoil permeability, μ_{rm}, nearly has a value from 1.03 to 1.1. More on recoil line and demagnetization characteristics is considered in later sections with worked examples.

1.1.2 Operating Point and Air Gap Line

To find the operating point on the demagnetization characteristic of the magnet, consider the flux path in the machine [6]. The flux crosses from the north pole of the rotor magnet to stator through an air gap and then closes the flux path from stator to rotor south pole, via air gap. In the process, the flux crosses two times the magnet length and two times the air gap, as shown in Figure 1.4. The mmf provided by magnets is equal to the mmf received by the air gap if the mmf requirement of stator and rotor iron is considered negligible. Then

$$H_m \ell_m + H_g \ell_g = 0 \tag{1.6}$$

where

H_m and H_g are magnetic field intensities or strengths in magnet and air, respectively

ℓ_m and ℓ_g are the length of the magnet and air gap, respectively

The operating flux density on the demagnetization characteristic can be written assuming that it is a straight line as

$$B_m = B_r + \mu_0\mu_{rm} H_m \tag{1.7}$$

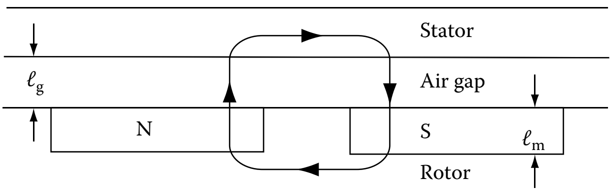


FIGURE 1.4 Simple layout of the stator and rotor of a machine. (From Krishnan, R., *Electric Motor Drives*, Figure 9.3, Prentice Hall, Upper Saddle River, NJ, 2001. With permission.)

The air gap flux density in terms of its magnetic field density is

$$B_g = \mu_0 H_g \tag{1.8}$$

Substituting for H_m from Equation 1.6 into Equation 1.7 and then substituting for H_g in terms of B_g from Equation 1.8 gives

$$B_m = B_r + \mu_0\mu_{rm}H_m = B_r - \mu_0\mu_{rm}\frac{H_g\ell_g}{\ell_m} = B_r - \mu_{rm}\frac{\ell_g}{\ell_m}\mu_o H_g = B_r - \mu_{rm}\frac{\ell_g}{\ell_m}B_g \tag{1.9}$$

Neglecting leakage flux leads to

$$B_g = B_m \tag{1.10}$$

Substituting Equation 1.10 into Equation 1.9, the operating magnet flux density is derived as

$$B_m = \frac{B_r}{\left(1 + \dfrac{\mu_{rm}\ell_g}{\ell_m}\right)} \tag{1.11}$$

This clearly indicates that the operating flux density is always less than the remanent flux density because of the air gap excitation requirement. Note that the excitation requirements of iron and leakage fluxes are neglected in this conceptual derivation. The equation gives design guidelines. Consider an air gap flux density that is equal to remanent flux density of the magnet. In this case, the denominator of the equation has to be the one which amounts to thickness of the magnet being much higher than the product of the air gap length and relative permeability of the magnet. Approximating the relative permeability of a high-grade magnet to be 1, the thickness of the magnet has to be much larger than air gap length to equal the air gap flux density to magnet's remanent flux density. This will result in a large magnet, which is not feasible in a practical machine from viewpoints of cost and compactness of the rotor and hence the machine overall. Assuming the thickness of the magnets is far higher than that of the air gap length, which is not practical, there is another issue that makes it impractical. There will be leakage fluxes between the magnets in the air gap without connecting the stator part of the machine and it will be significant when the gap between adjacent magnets is smaller compared to the thickness of the magnets. Therefore, the practical ratio encountered between the thickness of the magnet and air gap length is from 1 to 20. The lower this ratio is, the volume of the magnet and cost will be lower and so also the power output and power density of the machine. The higher this ratio, the characteristics described in previously are higher. Increasing the ratio does not lead to proportional benefit in power output and beyond a certain ratio between them, the benefits peak due to factors such as leakage fluxes and increasing volume and weight of the rotor, resulting in lower power density of the machine.

The operating point obtained from Equation 1.11 is shown in Figure 1.3 and the line connecting the operating flux density, B_m, and origin is known as air gap line or also load line. The slope of this line is equal to negative of a fictitious permeance coefficient, μ_c, times the permeance of the air. If the stator is electrically excited to produce demagnetization, then the load line moves toward the left and parallel to the load line as shown in the figure. The operating flux density is further reduced from B_m. Note that the permeance coefficient is derived for an operating point defined by B_m and H_m as

$$B_m = B_r + \mu_0\mu_{rm}H_m = -\mu_0\mu_c H_m \tag{1.12}$$

where H_m is the magnet field intensity due to stator current excitation in the machine. The permeance coefficient is then derived as

$$\mu_c = -\frac{B_r}{\mu_o H_m} - \mu_{rm} \tag{1.13}$$

But the remanent flux density B_r can be expressed by assuming as though it is a function of operating magnetic field intensity, H_m as

$$B_r = -\mu_0\mu_{re}\left(H_m\right)H_m \tag{1.14}$$

where $\mu_{re}(H_m)$ can be considered as external permeability and dependent on H_m. Substituting Equation 1.14 into Equation 1.13 gives the permeance coefficient as

$$\mu_c = \mu_{re}\left(H_m\right) - \mu_{rm} \tag{1.15}$$

The variations in the remanent flux density due to temperature changes as well as the impact of the applied magnetic field intensity, both of which are induced by external operating conditions, are clearly seen from this formulation of μ_c. As demagnetizing field is introduced by external operating conditions, it is seen that the permeance coefficient will decrease as the external permeability also decreases for that operating point. In hard PMs, the external permeability is in the order of 1–10 in the nominal operating region. It may be arrived at as in the following. Consider an example with a high-grade PM having a remanent flux density of 1.2 T and corresponding coercivity of -9×10^6 A/m. Let the air gap line produce, say, a flux density of 0.8 T. To find the permeance coefficient, the relative permeability of the magnet has to be evaluated, which is obtained from

$$\mu_{rm} = -\frac{B_r}{\mu_o H_m} = -\frac{1.2}{4\pi\times10^{-7}(-0.9\times10^6)} = 1.058$$

The operating magnetic field intensity for the operating flux density of 0.8 T is

$$B_m = B_r + \mu_0\mu_{rm}H_m = 0.8 = 1.2 + \left(4\pi\times10^{-7}\right)1.058H_m$$

from which

$$H_m = -0.3\times10^6 \text{ A/m}$$

Using this value, the permeance coefficient can be calculated as

$$\mu_c = -\frac{B_r}{\mu_o H_m} - \mu_{rm} = 2.13$$

And likewise, for operating flux density of 1 T, the permeance coefficient is 5.32. As the operating flux density is lowered to zero, the permeance coefficient goes to zero and for higher flux densities, the permeance coefficient increases, and for realistic operating points up to 1.1 T (that is, nearly 92% of remanent flux density), it stays within a value of 10. Which brings up the point that for high-grade PMs, the permeance coefficient is in the range of 1–10 and hence it may be used as an indicator of the magnet's operating flux density.

1.1.3 Energy Density

The energy density of the magnet is found by taking the product of its magnetic field strength and operating flux density. This measure serves to differentiate the magnets for use in machines with those having higher values preferred for high-power density machines. The peak energy operating point is optimal from the point of view of magnet utilization. The maximum energy density, E_{max}, for a hard high-grade PM shown in Figure 1.5 is derived as follows. The energy density, E_m, in terms of the product of magnet flux density, B_m, and its field intensity, H_m, is found and then taking its derivative with respect to field intensity and equating it to zero gives the condition that leads to maximum energy density. The steps to find maximum energy density are

$$E_m = B_m H_m = \left(B_r + \mu_0\mu_{rm}H_m\right)H_m \tag{1.16}$$

Taking the derivative of this energy density as

$$\frac{dE_m}{dH_m} = 0 = B_r + 2\mu_o\mu_{rm}H_m \tag{1.17}$$

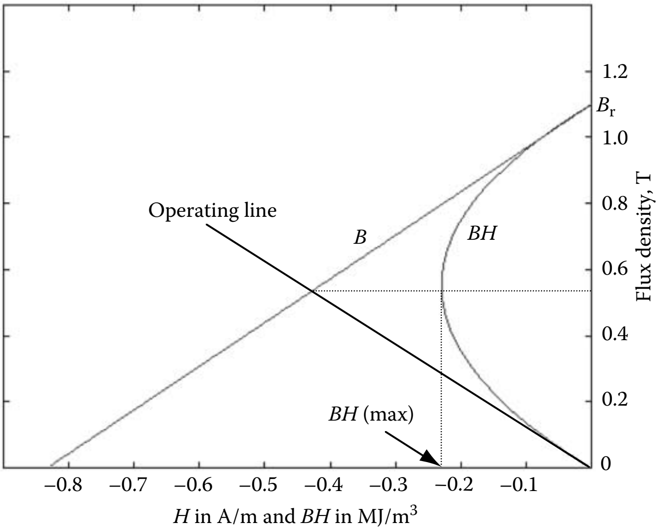


FIGURE 1.5 Energy product and preferred operating characteristics. (From Krishnan, R., *Electric Motor Drives*, Figure 9.4, Prentice Hall, Upper Saddle River, NJ, 2001. With permission.)

from which the field intensity is derived as

$$H_m = -\frac{B_r}{2\mu_o\mu_{rm}} \tag{1.18}$$

Substituting Equation 1.18 into Equation 1.16, the maximum energy density is derived as

$$E_{max} = -\frac{B_r^2}{4\mu_o\mu_{rm}} \tag{1.19}$$

Substituting Equation 1.18 into the expression for magnet flux density inserted in Equation 1.16, the flux density at which the maximum energy density available is found to be at half the remanent flux density, $0.5B_r$. The operating line for this flux density is shown in the figure giving the required magnetic field strength. Note that this operating point for maximum energy density requires a considerable amount of demagnetizing field strength from the stator excitation of the machine. Moreover, it is not practical to maintain this operating point in a variable speed machine drive as the stator currents will be varying widely over the entire torque speed region. To read the energy density in Figure 1.5, the projection of *BH* curve on *H*-axis is considered as the energy has to be negative. Some authors plot it in quadrant III to avoid confusion in this regard but here it is plotted in the second quadrant itself to keep the figure small and compact.

For a low-grade PM such as alnico, the magnet flux density and energy density characteristics are shown in Figure 1.6. The flux density corresponding to maximum

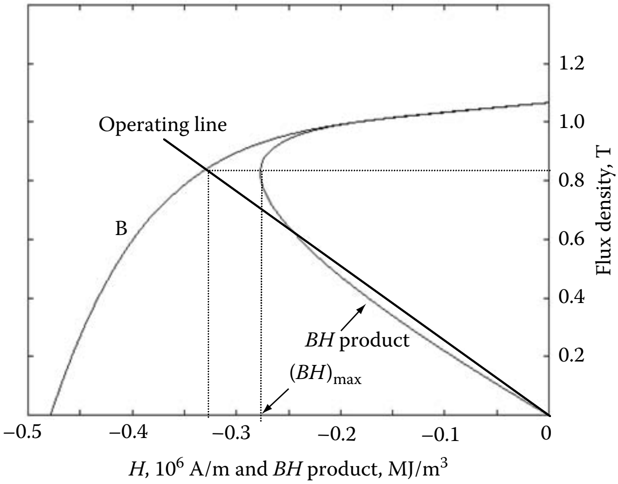


FIGURE 1.6 The energy density characteristic of an alnico magnet.

energy density is higher than half the remanent flux density in contrast to the high-grade PM characteristics. The maximum energy density is around one-tenth of the high-grade magnet's energy density is to be observed. Therefore, for high-performance applications requiring small motor weight and volume, these magnets are not preferred for use in electromagnetic devices.

1.1.4 Energy Stored in the Magnet

Energy density gives an indication of where best to operate the magnet in a device but it is not the same as the energy stored in the magnet. From fundamental electromechanics, the energy stored in a device is given by

$$W_m = (\text{Volume})\int H\,dB \tag{1.20}$$

Or energy stored per unit volume is given as

$$\frac{W_m}{\text{Volume}} = \int H\,dB \tag{1.21}$$

For example, consider a high-grade magnet with linear demagnetization characteristic and for that the energy stored per unit volume is

$$\frac{W_m}{\text{Volume}} = \int_0^{B_r} H\,dB = \int_0^{B_r}\left(\frac{B-B_r}{\mu_o\mu_{rm}}\right)dB = -\frac{B_r^2}{2\mu_o\mu_{rm}} = 2\left[(BH)_{max}\right] = 2E_{max}\ (\text{J/m}^3) \tag{1.22}$$

$(BH)_{max}$ is nothing but the maximum energy density derived in Equation 1.19 and denoted as E_{max}.

1.1.5 Magnet Volume

The volume of magnet embedded in the rotor determines the single most important cost component of the PM electrical machine, particularly in the case of the high power density magnet machines. It is critical in all applications to minimize the volume of the magnets among other things to minimize the cost of the machine as well as to have a smaller and lighter rotor. The magnet volume is found in terms of operating energy density and air gap volume as follows. Using Equation 1.6 and substituting for H_g in terms of B_g

$$B_g \ell_g = -\mu_0 H_m \ell_m \tag{1.23}$$

$$B_m A_m = B_g A_g \tag{1.24}$$

Using these ideal relationships, the magnet volume is derived as

$$V_m = A_m \ell_m = -\left(\frac{B_g A_g}{B_m}\right)\left(\frac{B_g \ell_g}{\mu_0 H_m}\right) = \frac{B_g^2 \left(A_g \ell_g\right)}{\mu_0 \left|B_m H_m\right|} = \frac{B_g^2 V_g}{\mu_0 \left|E_m\right|} \tag{1.25}$$

where

V_g is the air gap volume
E_m is the magnet operating energy density
A_m and A_g are the magnet and air gap area

Note that the equation is framed in absolute magnitude of the energy density by factoring $-H_m$ as a positive value with an absolute sign. From this relationship, it is inferred that the maximum operating energy density point of the magnet will yield the magnet with minimum volume and hence minimum cost. This is a simplified relationship since the leakage flux between the magnets is ignored and even then the result is not far off from reality for preliminary design calculation.

1.1.6 Effect of External Magnetic Field Intensity

PMSMs experience a magnetic field due to the excitation of its armature windings in the stator in addition to the field due to PMs on the rotor. The interaction of these fields has a certain consequence depending on whether the fields are in the same or opposite directions. The external excitation (due to the winding excitation) is always intended for weakening the flux in the air gap. Therefore, the flux due to external excitation is directed in the opposite direction to that of the PM rotor flux. It is not usual to cumulatively compound the fluxes as it would strengthen the air gap flux and hence will result in the saturation of the stator core laminations and consequently in higher core losses. The reason for not allowing to cumulative compounding of fluxes

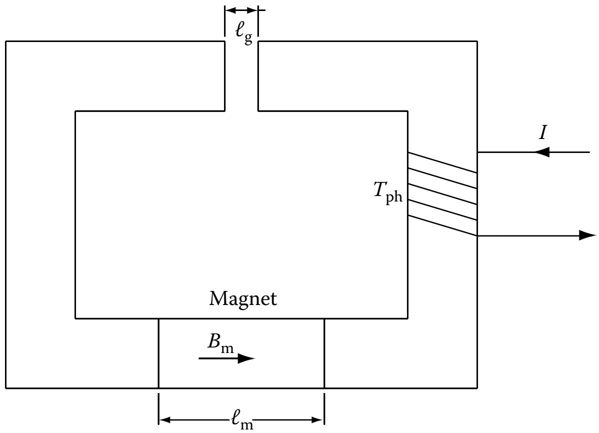


FIGURE 1.7 Simple electromagnetic device with a PM.

is that during design phase of the electrical machine, the magnet flux alone is set to provide the operating point near the knee point in the B–H (or equivalently flux versus current) characteristics of the lamination materials. Any further compounding of the flux with external excitation will move the operating point into saturation. Consider a simple electromagnetic device shown in Figure 1.7 to illustrate the effect of external magnetic field intensity. The coil has T_{ph} turns and carries a current of I. Assume that the external current is zero and hence only the field due to the PM exists. That would circulate a flux in the core in the anticlockwise direction as the magnet polarity is defined to be from left to right. If the winding is excited with a current, then the flux produced by it will traverse the core in the same direction as that of the PM flux. That would increase the flux in the core and in the air gap as the two fluxes due to the winding excitation and PM field are additive. If the current direction in the winding is opposite to that shown in figure, then the PM flux will be opposed by the external field due to the winding flux, resulting in the net reduction of the flux in the core and air gap. Two approaches, analytical and graphical, are developed in the following for the determination of the operating load flux density in the magnet when the electromagnetic device is superposed with an external excitation through a winding.

1.1.6.1 Analytical Approach

Let the magnet operating flux density is given by

$$B_m = B_r + \mu_o \mu_{rm} H \tag{1.26}$$

and the mmf of the external source must be equal to that of the mmf requirement of the air gap and magnet. The mmf required for the iron path is neglected here to get a conceptual result. The mmf around the flux path then is written as

$$H_m \ell_m + H_g \ell_g = T_{ph} I \tag{1.27}$$

where the subscript m and g correspond to variables in magnet and air gap, and ℓ corresponds to length of the air gap and magnet with respective subscripts. The magnetic field intensity is then

$$H_{\mathrm{m}} = \frac{T_{\mathrm{ph}} I - H_{\mathrm{g}} \ell_{\mathrm{g}}}{\ell_{\mathrm{m}}} \tag{1.28}$$

But the magnetic field intensity in the air gap is related to the air gap flux density as

$$B_{\mathrm{g}} = \mu_0 H_{\mathrm{g}} \tag{1.29}$$

and there is a clear relationship between the flux densities in the air gap and magnet and it can be modeled as follows. The useful flux, which is in the air gap, is always a fraction of the magnet flux, which is expressed in terms of the constant of proportionality, $1/k$, also known as leakage factor:

$$\phi_{\mathrm{g}} = \frac{\phi_{\mathrm{m}}}{k} \tag{1.30}$$

But the air gap and magnet flux in terms of the flux densities and areas of cross section are

$$\phi_{\mathrm{g}} = B_{\mathrm{g}} A_{\mathrm{g}}; \quad \phi_{\mathrm{m}} = B_{\mathrm{m}} A_{\mathrm{m}} \tag{1.31}$$

where the area of cross-section of the air gap and magnet are given as A_{g} and A_{m}, respectively. Thus defining the flux density of the air gap in terms of the magnet flux density by combining the equations from Equation 1.27 to Equation 1.31 and substituting in the magnet flux density equation of the magnet (Equation 1.26), the following is obtained:

$$B_{\mathrm{m}} = \frac{1}{1 + \mu_{\mathrm{rm}} \dfrac{\ell_{\mathrm{g}}}{\ell_{\mathrm{m}}} \dfrac{A_{\mathrm{m}}}{A_{\mathrm{g}}} \dfrac{1}{k}} \left[B_{\mathrm{r}} + \mu_0 \mu_{\mathrm{rm}} \frac{T_{\mathrm{ph}} I}{\ell_{\mathrm{m}}} \right] \tag{1.32}$$

In the case of a high-grade PM, note that the remanent flux density is directly proportional to coercive force as the demagnetization characteristic is a straight line and can be expressed as

$$B_{\mathrm{r}} = -\mu_{\mathrm{o}} \mu_{\mathrm{rm}} H_{\mathrm{c}} q \quad \text{or alternately} \quad q B_{\mathrm{r}} = \mu_{\mathrm{o}} \mu_{\mathrm{rm}} \left| H_{\mathrm{c}} \right| \tag{1.33}$$

In the case of low-grade PM such as alnico, the same relationship can be used but instead of the real coercive force, a fictitious coercive force is considered for modeling. It is obtained by extending the operating line until it intersects the field intensity

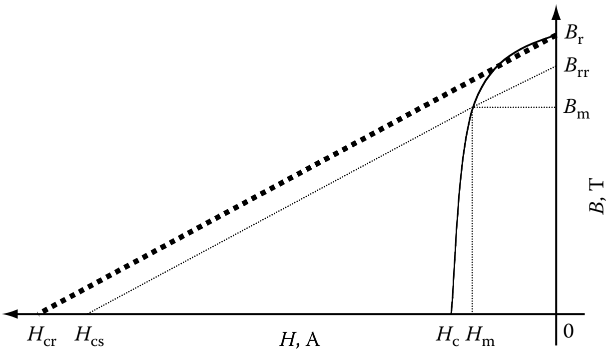


FIGURE 1.8 The alnico magnet's fictitious coercive force for an operating point.

axis and the intersecting field intensity, H_{cs} for an operating flux density of B_m and shown in Figure 1.8.

Realizing that the external mmf, the product of number of winding turns and current, over the magnet length is the external field intensity, H_e, yields the flux density equation in such terms as

$$B_m = \frac{B_r}{1+\mu_{rm}\dfrac{\ell_g}{\ell_m}\dfrac{A_m}{A_g}\dfrac{1}{k}}\left[1+\mu_0\mu_{rm}\frac{H_e}{B_r}\right] \tag{1.34}$$

where

$$H_e = \frac{T_{ph}I}{\ell_m} \tag{1.35}$$

Substituting the remanent flux density in terms of the coercive force (real in the case of high-grade and fictitious in the case of the low-grade PMs) given in Equation 1.33 in the flux density equation (Equation 1.34) gives

$$B_m = \frac{B_r}{1+\mu_{rm}\dfrac{\ell_g}{\ell_m}\dfrac{A_m}{A_g}\dfrac{1}{k}}\left[1-\frac{H_e}{H_c}\right] = B_0\left[1+\frac{H_e}{|H_c|}\right] \tag{1.36}$$

where

$$B_o = \frac{B_r}{1+\mu_{rm}\dfrac{\ell_g}{\ell_m}\dfrac{A_m}{A_g}\dfrac{1}{k}} \tag{1.37}$$

where B_0 is the no load flux density, i.e., with no externally applied field intensity. From Equation 1.36, it is inferred that the external excitation reduces the flux density if the excitation is against the magnet flux. Reversing the excitation will increase the magnet flux but invariably lead to saturation of the iron core and therefore is not attempted in practice.

1.1.6.2 Graphical Approach

Instead of the analytical approach, the load flux density can also be found graphically. Such a procedure is derived here from the analytical expressions derived earlier. It is illustrated for the low-grade PM device.

The no load flux density is written as

$$B_0 = B_{rr} + \mu_o \mu_{rm} H_o = B_{rr}\left[1 - \frac{H_o}{H_{cs}}\right] \tag{1.38}$$

where subscript "o" corresponds to operating recoil line and as shown in Figure 1.9. The remanent flux density under the new operating condition is B_{rr} and corresponding coercivity is H_{cs}, which is fictitious for low-grade magnet such as the one shown in Figure 1.8. Similarly the load flux density is derived by substituting for B_o from Equation 1.37 into Equation 1.38 as

$$B_m = B_o\left[1 - \frac{H_e}{H_{cs}}\right] = B_{rr}\left[1 - \frac{H_o}{H_{cs}}\right]\left[1 - \frac{H_e}{H_{cs}}\right] = B_{rr} + \mu_o \mu_{rm} H_m \tag{1.39}$$

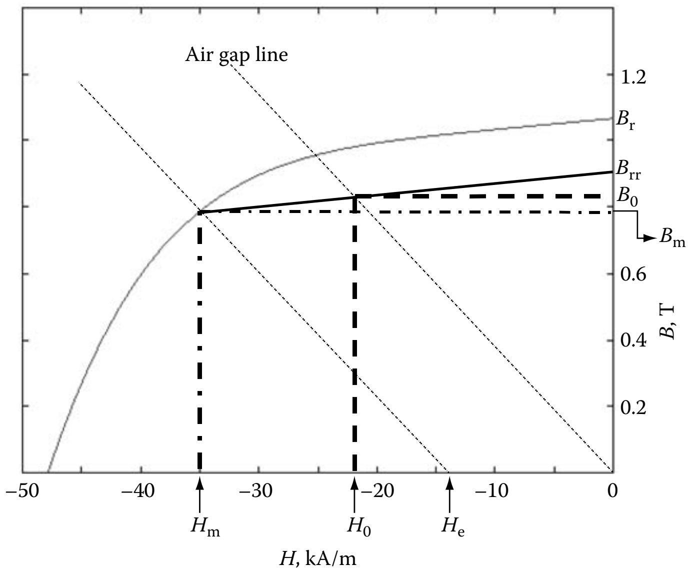


FIGURE 1.9 The magnet characteristic and operating points with and without external excitations.

On rearranging this equation to obtain the magnetic field intensity at the operating point as

$$H_m = \frac{B_{rr}}{\mu_o \mu_{rm}}\left[-\frac{H_o + H_e}{H_{cs}} + \frac{H_o H_e}{H_{cs}^2}\right] = -H_{cs}\left[-\frac{H_o + H_e}{H_{cs}} + \frac{H_o H_e}{H_{cs}^2}\right] = H_o + H_e - \frac{H_e}{H_{cs}} H_o$$

$$= H_o + H_e + \left[\frac{B_m}{B_o} - 1\right] H_o = H_e + \left[\frac{B_m}{B_o}\right] H_o \tag{1.40}$$

Equation 1.40 is rearranged in a manner to bring out explicitly the equality of air gap line and load line with an external excitation of H_e as

$$\frac{H_m - H_e}{B_m} = \left[\frac{H_o}{B_o}\right] \tag{1.41}$$

This demonstrates that the operating flux density, B_m, is given by the line that is parallel to the air gap line that is shifted from the origin by the external magnetic field intensity generated from the excitation of the coil. Its construction is shown in Figure 1.9.

Example 1.1

A PM whose characteristics are given in Figure 1.9 is embedded in the electromagnetic device shown in Figure 1.6. The magnet is stabilized at −35 kA/m with a new remanent flux density. The relative permeability of iron is considered very large so the reluctance of the iron is zero to simplify calculations. Find the current in the coil required to operate the magnet at its stabilized operating point. The data for the device are

$\ell_g = 0.5$ mm
$\ell_m = 12$ mm
Leakage factor, $k = 1.25$
Magnet area = air gap area
Number of turns in the coil, $T_{ph} = 1000$

Solution

The approach to solve this problem is to find the magnet permeability from the characteristic given, then to proceed to find the new remanent flux density. Given the remanent flux density, the magnet flux density with no external excitation is found, which gives the air gap line. From this, the solution can be obtained graphically by drawing a line parallel to the air gap line that is shifted to the left of the origin by an amount equal to externally excited field intensity from the stabilized operating point. The intersection of this line on the field strength axis gives the external excitation required to demagnetize the magnet. Graphical solution is shown in Figure 1.9. Alternatively, it is evaluated analytically as in the following.

Magnet relative permeability is found from the characteristic by considering the remanent flux density and another operating point B_1 at H_1 as

$$\mu_{rm} = \frac{B_r - B_1}{\mu_o H_1} = \frac{1.0665 - 1.033}{(4\pi \times 10^{-7}) \times (10 \times 10^3)} = 2.665$$

As the magnet is stabilized at $H_d = -35$ kA/m, which gives the magnet operating point B_m (0.7856 T), the new remanent flux density is found by drawing the recoil line, which is a line parallel to the magnet characteristic from this operating point or analytically obtained from

$$B_m = B_{rr} + \mu_o \mu_{rm} H_m$$

and therefore the new remanent flux density (for the stabilized operating point) is

$$B_{rr} = B_m - \mu_o \mu_{rm} H_m = 0.7856 - 4\pi \times 10^{-7} \times 2.665 \times (-35{,}000) = 0.9028 \text{ T}$$

The operating point of the magnet with no external excitation in the device is obtained from

$$B_o = \frac{B_{rr}}{1 + \mu_{rm} \frac{l_g}{l_m} \frac{A_m}{A_g} \frac{1}{k}} = \frac{0.9028}{1 + 2.665 \times \frac{0.5}{12} \times 1 \times \frac{1}{1.25}} = 0.8291 \text{ T}$$

Fictitious coercivity of the magnet in low-grade magnets is

$$H_{ce} = \frac{B_{rr}}{\mu_o \mu_{rm}} = 269.5 \text{ kA/m}$$

The flux density lowered by external excitation is

$$\Delta B = \frac{B_m - B_o}{B_o} = \frac{0.7856 - 0.8291}{0.8291} = -0.05246 \text{ T}$$

The external field strength required to demagnetize the magnet is

$$H_e = -\Delta B \left| H_{ce} \right| = -0.5246 \times 269.5 \times 10^3 = -14.14 \times 10^3 \text{ A/m}$$

Mmf to create this field strength is given by

$$F_e = H_e \ell_m = -14.14 \times 10^3 \times 0.012 = -169.68 \text{ AT}$$

The demagnetizing current is obtained from it as

$$I = \frac{F_e}{T_{ph}} = \frac{-169.68}{1000} = -0.16968 \text{ A}$$

Example 1.2

The electromagnetic device shown in Figure 1.6 is considered for this problem. The magnet is of the neodymium variety with the remanent flux density of 1.1 T and coercivity of 0.86×10^6 A/m and has a straight line demagnetization characteristic. Determine the flux density in the air gap when the current in the coil is 10 A. The data for the device are given in Example 1.1.

Solution

The solution is very similar to the solution for Example 1.1. The graphical solution is shown in Figure 1.10.

Or analytically the solution is found as follows:

$$B_m = B_r + \mu_o \mu_{rm} H$$

and at $B = 0$, $H = H_c = 0.87 \times 10^6$ A/m, which on substitution gives the relative permeability of the magnet as

$$\mu_{rm} = \frac{B_r}{\mu_o H_c} = \frac{1.1}{4\pi \times 10^{-7} \times 0.87 \times 10^6} = 1.0062$$

The no load operating flux density, B_o, is obtained from

$$B_o = \frac{B_r}{1 + \mu_{rm} \frac{\ell_g}{\ell_m} \frac{A_m}{A_g} \frac{1}{k}} = \frac{1.1}{1 + 1.0062 \times \frac{0.0005}{0.012} \times 1 \times \frac{1}{1.25}} = 1.0643 \text{ T}$$

The line connecting origin and the no load flux density gives the air gap line or no load line. To evaluate the load operating flux density, a line parallel to the air

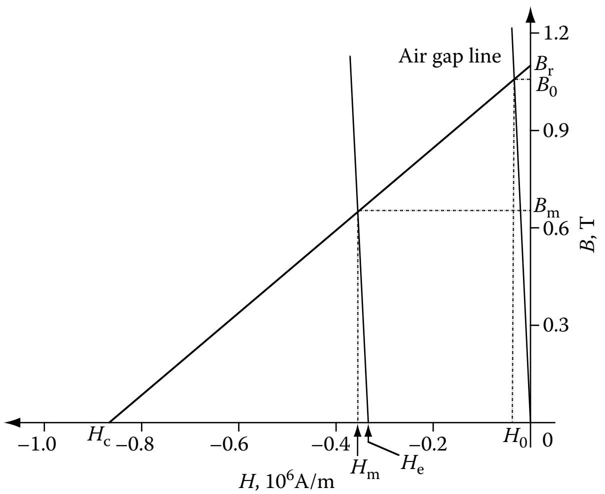


FIGURE 1.10 Operating point determination with external excitation for Example 1.2.

gap line shifted from the origin to the left by an amount equal to the external field strength (H_e) has to be drawn to intersect the demagnetization characteristic of the magnet. The external field strength is computed as follows:

The external excitation, $F_e = T_{ph}I = -1000 \times 4 = -4000$ AT

The external field strength

$$H_e = \frac{F_e}{\ell_m} = -\frac{4000}{0.012} = -0.333 \times 10^6 \text{ A/m}$$

Drawing the load line parallel to the air gap line from H_e intersects the magnet demagnetization characteristic yielding the magnet flux density B_m as 0.6575 T. Analytically the magnet load flux density is found from

$$B_m = \frac{1}{1 + \mu_{rm} \frac{\ell_g}{\ell_m} \frac{A_m}{A_g} \frac{1}{k}} \left[B_r + \mu_o \mu_{rm} \frac{T_{ph} I}{\ell_m} \right] = 0.6575 \text{ T}$$

Note that the air gap flux density is

$$B_g = \frac{B_m A_m}{k A_g} = \frac{0.6575}{1.25} = 0.526 \text{ T}$$

1.2 ARRANGEMENT OF PMs

The magnets can be individually made in many shapes and sizes. They also come in the ring form. The ring form is the easiest to install as they can be slided on top of the steel laminated rotor and held in place by suitable means. Then they can be magnetized in any desired direction. The disadvantage of the ring magnets is that they are expensive compared to individual magnets. The individual magnets can take any shape. Each pole may consist of a number of magnet segments instead of one piece. The reason for this is that for large machines with constraints on machining of the magnet materials and charging them to magnetize, the use of a single magnet piece or segment per pole may not be possible or if possible it may not be the most economical. In some cases, a single magnet per pole is not suitable for large flux weakening range of operation. It is found that the flux weakening capability is enhanced with multisegment magnet per pole structures. They can also be stacked one on top of another with equal or varying widths to yield a desired flux distribution such as a sinusoidal or trapezoidal distribution in the air gap of the PM synchronous and brushless dc machines, respectively. The three kinds of magnet poles are shown in Figure 1.11. The magnets may have varying thicknesses and widths for stacking up and note that the magnetization polarities are in the same direction for the stack.

All these arrangements have advantages and applications. Single magnet per pole is ideal for small machines. Higher power machines may use multisegment magnets

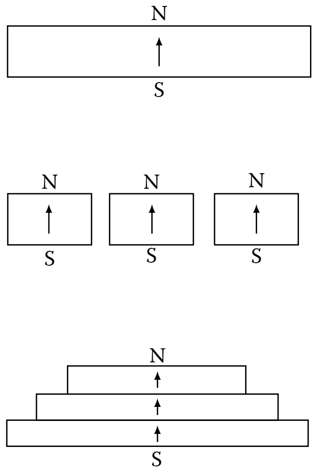


FIGURE 1.11 Realization of magnet poles with one or multiple segments.

per pole and stacked pole structures. Single magnet and multisegment magnets are prevalent in practice.

Various shapes of magnets that are in practice such as rectangular, radial, and breadloaf are shown in Figure 1.12. The radial and breadloaf types of magnets are ideal for surface mount-type of PMSMs. The air gap in the case of the radial magnets on the rotor surface is uniform, assuming that it is not partially embedded in the rotor lamination with the result that the air gap flux density is uniform. Radial magnets can come with their end surfaces being radial or parallel and accordingly they are referred to as surface radial and surface parallel magnets and they are shown in Figure 1.13. The air gap for the breadloaf magnet is nonuniform and hence the air gap flux density also being nonuniform. That allows for shaping the flux density in the air gap, which can be other than rectangular and constant over the magnet arc. This is elaborated in Section 1.3.

Rectangular magnets are commonly used in interior PM rotor construction. The recent trend is to have multiple magnets per pole in this construction to facilitate higher flux-weakening region of operation. This structure is not ideal in surface

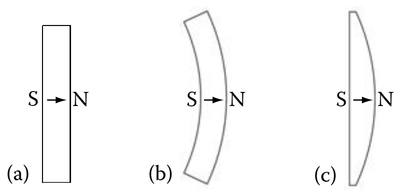


FIGURE 1.12 Various types of magnets: (a) rectangular; (b) radial; and (c) breadloaf.

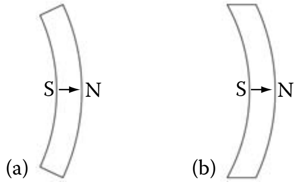


FIGURE 1.13 Radial magnets with radial and parallel surfaces. (a) Surface radial magnet and (b) surface parallel magnet.

mount machines as the desired constant air gap cannot be maintained. Other shapes are possible and open to the imagination of the designer but may be limited by manufacturing cost and techniques. The constraining factors in regard to shapes are as follows:

Minimum material per magnet pole
Ease of machining and construction
Flux density distribution that is rectangular or sinusoidal

Because of these constraints, many other shapes have not seen the light of the day.

1.3 MAGNETIZATION OF PMs

PMs are magnetized [56–59] with certain orientation or direction such as radial, parallel, or any other direction. The magnetization orientation strongly influences the quality of the air gap flux density distribution and indirectly affects the power density in a given arrangement of the machine with PMs. The air gap flux density distribution influences in turn the harmonic torque generation in the machine and the presence of the harmonic torques corrodes the quality of torque output in the machine, particularly in the case of high-performance motor drives. Radial and parallel magnetization are prevalent in practice whereas other forms of magnetization are yet to make their presence felt even when they have been known to possess unique advantages in some cases.

1.3.1 Radial and Parallel Magnetizations

Radial and parallel magnetization are shown in Figure 1.14. The normal direction to the surface is indicated by vector **n** and the magnetization vector by **M**. The radial magnetization is along the radius while the parallel magnetization is parallel to the edges as in the case of surface parallel magnets. But that need not be true in cases with other edge shapes. The key in that case is that the magnetization direction is horizontal as shown in the figure. The effect of two such magnetizations in a synchronous machine is shown in the form of air gap flux density versus rotor position using finite element analysis as well their flux plots in Figure 1.15 for a machine

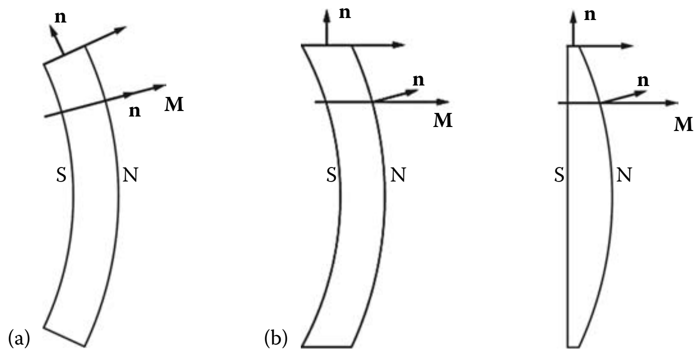
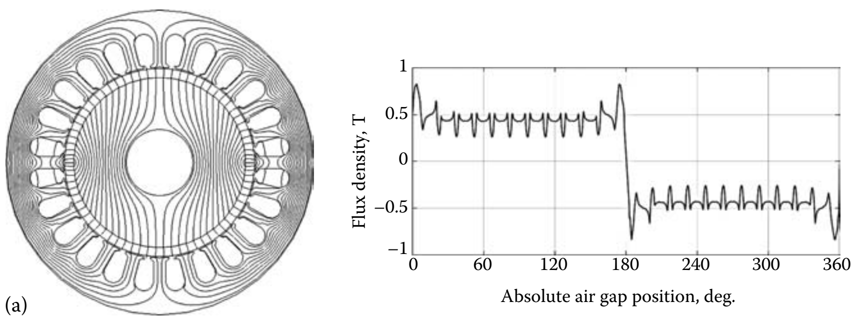


FIGURE 1.14 Magnetization of magnets. (a) Radial magnetization and (b) parallel magnetization



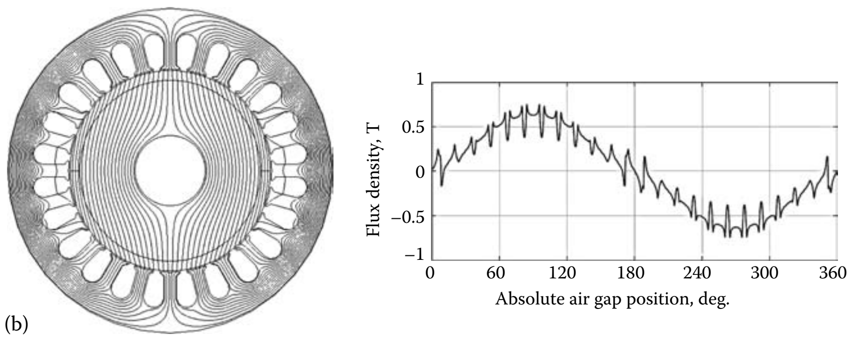


FIGURE 1.15 Flux and flux density versus rotor position for radially and parallel magnetized magnet machines. (a) Radially magnetized magnet machine. (b) Parallel magnetized magnet machine.

with 24 slots and 2 poles. The radially magnetized magnets produce a rectangular flux density distribution in the air gap whereas the parallel magnetized magnets produce a sinusoidal air gap flux density distribution. The reasons for the different flux density distributions can be explained from the following. The magnet flux leaves the PM and enters the stator in normal direction to their surfaces. The flux density vector in the radially magnetized magnets is normal along the radial direction itself, resulting in the maximum flux flowing in the radial direction also. Therefore, the air gap flux and its density are maximum and remain uniformly constant across the magnet. In the case of the parallel magnetized magnets, the normal component of the flux density vector that enters the stator can be seen to be proportional to sine of the angle between the x-axis and the magnetization vector M at that point in the magnet with the result that the flux density distribution is sinusoidal in the air gap. There is also a tangential component of the flux density in the case of the parallel magnetized magnets embedded machine while it is zero in the radially magnetized magnet machine.

1.3.2 Halbach Array

There is also the case where the magnets need not be magnetized in radial or parallel directions but can be in any direction. Such magnetization is exploited in an arrangement due to Halbach. Radial or parallel magnetization requires a back iron for better utilization of the magnets and these are the cases that have been illustrated so far. They are also the most prevalent in practice since they are much easier to realize. Before the Halbach arrangement is considered, the radial magnet arrangement is considered again here for a plain magnet arrangement (or an array) in air and its flux paths shown in Figure 1.16. Note that the magnets are of alternate polarity. To maximize magnet utilization, iron is required on both the top and the bottom of the magnet arrangement. In a machine, one side of the magnet array faces the stator through the air gap and the other side sits on the rotor lamination, and both the stator and rotor have steel laminations.

The Halbach arrangement [60–69] (sometimes known as Halbach array) shown in Figure 1.17 is a combination of two magnet arrays: one radial magnet array and one azimuthal magnet array. The combination is made in such a way that the succeeding magnets in the combined or Halbach array have either a clockwise or counterclockwise magnetization. Its flux distribution can be arrived at by summing the

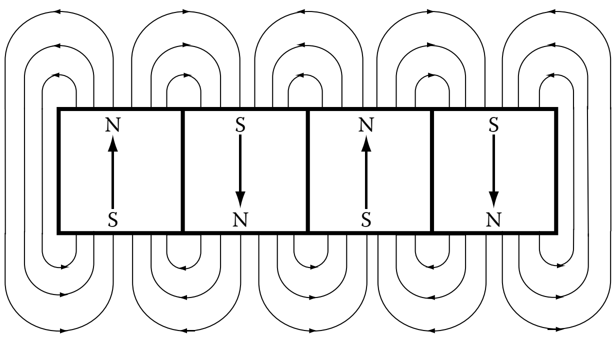


FIGURE 1.16 Radial magnet array in air.